

Undergraduate Course Catalogue

Faculty of Engineering - Bachelor of Engineering

Eurisko University, Beirut, Lebanon

Catalogue year 2026-2027

Two programmes | 33 courses | 72 credits per degree

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The Bachelor of Engineering

The Faculty of Engineering at Eurisko University offers two undergraduate programmes leading to the degree of Bachelor of Engineering: Computer Engineering and Mechanical Engineering. Each is a four-year, eight-term course of study comprising 72 credits, including a supervised industrial placement and a senior design project assessed by a panel drawn from industry.

Both programmes share a common foundation in mathematics, the physical sciences and computing, taken over the first two years, and diverge from the third year into discipline-specific study. A normal full-time load is 9 to 12 credits per term.

This Catalogue sets out degree requirements and course descriptions. Rules on grading, progression, fees and conduct are published separately in the Student Handbook.

How degree requirements work

Requirements are expressed as credits within five requirement categories rather than as a single list of courses. A student must satisfy every category individually. Credits earned beyond the requirement of one category do not count toward a shortfall in another, so a student may hold well over half the total credits and still be several terms from graduating.

Category	Credits	What it covers
Engineering Core	28	The ten mathematics, science and computing courses taken by every engineering student. All ten are required.
General Education	9	9 credits chosen from the four General Education courses. Any three satisfy the requirement.
Major Core	18	The six courses specific to the student's programme. All six are required.
Professional Practice and Capstone	8	Engineering economics, the industrial placement, and the senior design project. All three are required.
Technical Electives	9	9 credits chosen from the four ENGR 450-series electives. Any three satisfy the requirement.

Each course lists its prerequisites. Where more than one is listed, all are required, and each must have been passed at a grade of C- or above before the dependent course may be taken.

Programmes

Computer Engineering (BE-CENG)

Bachelor of Engineering (B.E.) | 4 years | 72 credits

Category	Credits	Courses
Engineering Core	28	ENGR 100, MATH 101, MATH 102, MATH 201, MATH 202, MATH 301, PHYS 101, PHYS 102, CHEM 101, CMPS 101
General Education	9	9 credits from: ENGL 101, ENGL 201, HUMN 210, SOCI 240
Computer Engineering Major Core	18	CMPS 102, CENG 310, CENG 320, CENG 330, CENG 410, CENG 420
Professional Practice and Capstone	8	ENGR 230, ENGR 480, ENGR 490
Technical Electives	9	9 credits from: ENGR 450, ENGR 451, ENGR 452, ENGR 453

Mechanical Engineering (BE-MECH)

Bachelor of Engineering (B.E.) | 4 years | 72 credits

Category	Credits	Courses
Engineering Core	28	ENGR 100, MATH 101, MATH 102, MATH 201, MATH 202, MATH 301, PHYS 101, PHYS 102, CHEM 101, CMPS 101
General Education	9	9 credits from: ENGL 101, ENGL 201, HUMN 210, SOCI 240
Mechanical Engineering Major Core	18	MECH 200, MECH 210, MECH 220, MECH 310, MECH 330, MECH 340
Professional Practice and Capstone	8	ENGR 230, ENGR 480, ENGR 490
Technical Electives	9	9 credits from: ENGR 450, ENGR 451, ENGR 452, ENGR 453

Course Descriptions

Courses are listed by subject prefix, then by number. Each entry gives the credit value and the prerequisites. Not every course is offered in every term; the schedule of classes for a given term is published separately by the Office of the Registrar.

Engineering (ENGR)

ENGR 100 Introduction to Engineering (1 credit)

Orientation to the engineering disciplines offered at Eurisko University, professional practice and licensure, engineering ethics, and expectations for academic work.

Prerequisite: none

ENGR 230 Engineering Economics (3 credits)

Time value of money, cash flow analysis, depreciation, and economic decision making under uncertainty.

Prerequisite: MATH 101

ENGR 450 Introduction to Artificial Intelligence (3 credits)

Search and optimisation, supervised and unsupervised learning, neural network fundamentals, and the responsible deployment of AI in engineering.

Prerequisites: CMPS 101, MATH 301

ENGR 451 Renewable Energy Systems (3 credits)

Solar photovoltaic and thermal systems, wind energy conversion, storage, and the techno-economic assessment of renewable installations.

Prerequisite: PHYS 102

ENGR 452 Robotics (3 credits)

Rigid body transformations, forward and inverse kinematics, trajectory planning, actuation and sensing.

Prerequisite: PHYS 102

ENGR 453 Data Analytics for Engineers (3 credits)

Data acquisition and cleaning, exploratory analysis and visualisation, regression and classification, and communicating quantitative findings.

Prerequisite: MATH 301

ENGR 480 Internship and Professional Practice (2 credits)

A supervised industrial placement of not fewer than 200 hours, documented in a placement report. Graded Pass/Fail.

Prerequisite: ENGR 230

ENGR 490 Senior Design Project (3 credits)

The capstone project: problem definition, concept selection, detailed design, prototyping or simulation, testing, and a final presentation to an industry panel.

Prerequisites: ENGL 201, ENGR 230

Mathematics (MATH)

MATH 101 Calculus I (3 credits)

Limits and continuity, differentiation, applications of the derivative, and an introduction to the definite integral.

Prerequisite: none

MATH 102 Calculus II (3 credits)

Techniques and applications of integration, sequences and series, and power series.

Prerequisite: MATH 101

MATH 201 Linear Algebra (3 credits)

Systems of linear equations, matrix algebra, vector spaces, eigenvalues and eigenvectors, with engineering applications.

Prerequisite: MATH 101

MATH 202 Differential Equations (3 credits)

First and second order ordinary differential equations, linear systems and Laplace transforms, applied to mechanical and electrical systems.

Prerequisite: MATH 102

MATH 301 Probability and Statistics for Engineers (3 credits)

Probability models, random variables and distributions, estimation, hypothesis testing and regression.

Prerequisite: MATH 102

Physics (PHYS)**PHYS 101 Physics I: Mechanics (3 credits)**

Kinematics, Newton's laws, work and energy, momentum and rotational motion. Includes a weekly laboratory.

Prerequisite: none

PHYS 102 Physics II: Electricity and Magnetism (3 credits)

Electrostatics, DC circuits, magnetic fields and electromagnetic induction. Includes a weekly laboratory.

Prerequisite: PHYS 101

Chemistry (CHEM)**CHEM 101 General Chemistry for Engineers (3 credits)**

Atomic structure and bonding, stoichiometry, thermochemistry and equilibrium, with emphasis on engineering materials and corrosion.

Prerequisite: none

Computer Science (CMPS)**CMPS 101 Introduction to Programming (3 credits)**

Algorithmic thinking, control flow, functions, arrays and files, and basic object-oriented concepts, taught in Python.

Prerequisite: none

CMPS 102 Data Structures and Algorithms (3 credits)

Lists, stacks, queues, trees, hash tables and graphs; sorting and searching; and asymptotic analysis of time and space complexity.

Prerequisite: CMPS 101

Computer Engineering (CENG)**CENG 310 Digital Logic Design (3 credits)**

Number systems, Boolean algebra and minimisation, combinational and sequential logic, and finite state machines.

Prerequisite: PHYS 102

CENG 320 Computer Architecture (3 credits)

Instruction set architecture, datapath and control, pipelining, and the memory hierarchy.

Prerequisite: CENG 310

CENG 330 Operating Systems (3 credits)

Processes and threads, scheduling, synchronisation and deadlock, memory management and file systems.

Prerequisite: CMPS 102

CENG 410 Database Systems (3 credits)

The relational model, entity-relationship design and normalisation, SQL, indexing and query processing, and an introduction to non-relational stores.

Prerequisite: CMPS 102

CENG 420 Software Engineering (3 credits)

Requirements, architecture and design, version control, testing strategy and delivery practices, applied to a team project.

Prerequisite: CMPS 102

Mechanical Engineering (MECH)

MECH 200 Engineering Statics (3 credits)

Force systems, equilibrium of particles and rigid bodies, trusses and frames, centroids and friction.

Prerequisite: PHYS 101

MECH 210 Thermodynamics (3 credits)

Properties of pure substances, the first and second laws, entropy, and power and refrigeration cycles.

Prerequisites: PHYS 101, CHEM 101

MECH 220 Materials Science for Engineers (3 credits)

Crystal structure and defects, phase diagrams, mechanical behaviour, heat treatment, and the selection of engineering materials.

Prerequisite: CHEM 101

MECH 310 Fluid Mechanics (3 credits)

Fluid statics, control volume analysis, the Bernoulli and momentum equations, and viscous flow in pipes.

Prerequisite: MECH 210

MECH 330 Mechanics of Materials (3 credits)

Stress and strain, axial loading, torsion, bending and shear in beams, deflection and column buckling.

Prerequisite: MECH 200

MECH 340 Machine Design (3 credits)

Failure theories, fatigue, and the design and selection of shafts, fasteners, springs, bearings and gears.

Prerequisite: MECH 330

English (ENGL)

ENGL 101 Academic English (3 credits)

Academic reading and writing, source evaluation, argument structure and citation conventions.

Prerequisite: none

ENGL 201 Technical Writing and Communication (3 credits)

Technical reports, specifications and proposals; document design; and oral presentation of technical material.

Prerequisite: ENGL 101

Humanities (HUMN)

HUMN 210 Ethics in Engineering and Technology (3 credits)

Ethical frameworks applied to engineering practice: professional codes, risk and public safety, data ethics, and case studies of engineering failure.

Prerequisite: none

Social Sciences (SOCI)

SOCI 240 Society, Technology and Development (3 credits)

The social and economic consequences of technological change, infrastructure policy, and the regional context of engineering practice.

Prerequisite: none